Ganyu Xu

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Education

U of Waterloo, MASc in Electrical and Computer Engineering, GPA 94.6/100 — Sep 2023 – Aug 2025
Courses: public-key cryptography, applied cryptography, post-quantum cryptography

U of California, Berkeley, BA in Mathematics and Statistics, GPA: 3.464/4.0 — Sep 2015 – May 2019

Technical skills

Programming languages: C, Python 3, Rust, Bash, SQL

Cryptography: Post Quantum Cryptography, Public Key Infrastructure, Transport Layer Security

Low-level/system programming: Embedded systems, valgrind, gdb/lldb, constant-time programming

Experience

Research Associate, Open Quantum Safe – Waterloo, Ontario, Canada — Aug 2025 – Present

- Core maintainer of **liboqs**, an open-source C library for quantum-resistant cryptographic algorithms.
- Triaged and fixed security vulnerabilities such as out-of-bounds memory access (#2266), cache timing side channel (#2431), and incorrect AVX512VL SHA3 implementation (#2442)
- Maintained liboqs testing infrastructure on GitHub Actions: expanding test coverage, reducing test runtime, and improving test reliability
- Engaged an international community of cryptography researchers and open-source contributors through code reviews, GitHub discussions, and community calls

Senior data engineer, LeanTaaS Inc. – Santa Clara, CA, United States — Jul 2019 – Sep 2023

- Deployed and maintained distributed Apache Airflow cluster on AWS Elastic Container Service.
- Improved data pipeline observability and production outage response time w/ DataDog and PagerDuty
- Mentored junior developers through project design review, pair programming, and technical workshops

Projects

Post-quantum TLS on embedded devices — xuganyu96/embedded-pqtls

- Designed and implemented a post-quantum key encapsulation mechanism based on ML-KEM with an alternative CCA transformation, achieving IND-1CCA security and reducing decapsulation time by more than 50%
- Expanded WolfSSL's support for post-quantum PKI and TLS handshake. Benchmarked performance on Pi Pico 2
- Implemented KEMTLS on WolfSSL, including KEM-based certificate generation and authentication protocol

RustCrypto — github.com/RustCrypto

- Contributed to Marvin Attack mitigation in RustCrypto/RSA: ported random prime generation to constant-time heap-allocated integers and integrated Marvin test harness for timing variability detection

rustls — github.com/rustls/rustls

- Traced demo client panic to incorrect TLS termination where the server closes the TCP connection before sending `close_notify`

Apache Airflow — github.com/apache/airflow

- Built `AirflowFargateExecutor`, a task executor that launches ephemeral Fargate containers per task for cost-efficient parallelism
- Fixed improper exception handling in python multiprocessing that causes critical process to become zombie instead of exiting upon database disconnect
- Implemented user-specified database check retries in the CLI command `airflow db check`